WAZE USER CHURN ANALYSIS

Executive Summary | Three-Notebook Learning Project

Foundations of Data Science · Go Beyond the Numbers: Translate Data into Insights · The Power of Statistics

Project focus
Inspecting user activity data, exploring churn-related patterns, and testing whether average drive counts differ by device type.

Tools
Python, pandas, NumPy, Matplotlib, Seaborn, SciPy, and Jupyter Notebook.

Important data provenance note
The notebooks load `waze_dataset.csv` from the Coursera lab environment. They do not document a public download URL or an independent collection process. This report therefore describes it as course-provided Waze user data.

Prepared from the three provided notebooks. Results and methods below reflect the notebooks' recorded outputs; interpretations are descriptive and should not be treated as causal findings.

1. Executive overview

This project follows a progression from data inspection to exploratory analysis and statistical inference. Across the three notebooks, the dataset contains 14,999 user-level observations and 13 original columns describing app sessions, drives, navigation behavior, driving distance and duration, activity frequency, device type, and a churn/retention label.

The initial inspection identified 700 missing churn labels, or approximately 4.7% of all rows. Among the 14,299 rows with known labels, 2,536 were marked churned and 11,763 retained, corresponding to a churn rate of approximately 17.7%. The EDA notebook highlights wide variation and right-skew in several activity measures, while the statistical notebook finds no statistically significant difference in average drive counts between iPhone and Android users at the 5% significance level.

The practical takeaway is to investigate user engagement and changes in activity over time rather than assuming device type explains churn. These notebooks provide descriptive and introductory inferential analysis; they do not establish causality or deliver a validated churn-prediction model.

2. Data source, structure, and acquisition

Each notebook reads the same file, `waze_dataset.csv`, using pandas. The lab instructions indicate that the file is automatically made available in the Coursera environment. No public URL, data collection procedure, or independent scraping/API workflow is documented in the notebooks. For portfolio documentation, the accurate description is “course-provided Waze user data.”

Item	Details
Dataset size	14,999 rows × 13 original columns
Target / label	`label`: retained or churned; 700 labels are missing
Behavior variables	Sessions, drives, total sessions, favorite navigation counts, driven distance, driving duration, activity days, driving days
User context	ID, days after onboarding, device (iPhone or Android)
Tools	pandas, NumPy, Matplotlib, Seaborn, SciPy

3. Notebook 1 — Foundations of Data Science

Purpose. Inspect the dataset and establish an initial understanding of data types, missingness, churn labels, and activity metrics before deeper analysis.

Processing steps. The notebook imports pandas and NumPy, loads the CSV, reviews sample rows and DataFrame metadata, compares summary statistics for rows with missing versus non-missing labels, and counts missing labels by device. It then compares churned and retained groups using counts, percentages, medians, and device composition. Three derived metrics are added: kilometers per drive, kilometers per driving day, and drives per driving day.

Data quality observations. There are 700 missing labels (447 iPhone users and 253 Android users). The notebook compares these counts with the overall device distribution; this does not establish a strong device-specific missingness issue. Missing labels should remain “unknown” unless additional documentation supports another interpretation. Derived ratios also require special care when their denominator is zero.

Key result. Among labeled records, 2,536 users are churned (17.7%) and 11,763 are retained (82.3%). The median number of drives is 50 for churned users and 59 for retained users. This is a descriptive difference, not proof that drive frequency causes retention.

4. Notebook 2 — Go Beyond the Numbers: Translate Data into Insights

Purpose. Use exploratory data analysis (EDA) and visualization to investigate distributions, potential outliers, activity behavior, and churn patterns.

Processing steps. The notebook imports pandas, NumPy, Matplotlib, and Seaborn; inspects rows, shape, data types, and descriptive statistics; and plots distributions using histograms, box plots, and categorical charts. It compares activity days with driving days, examines retention by device, calculates kilometers per driving day, explores churn proportions by driving-day count, and calculates the percentage of total sessions occurring in the last month. Selected highly skewed numeric columns are capped at their 95th percentile to reduce the influence of extreme values.

Cleaning decision to disclose. Dividing distance by driving days creates infinite values when driving days is zero. The notebook replaces positive and negative infinity with zero. This avoids infinite values in subsequent summaries, but zero then represents both a computed zero and a formerly undefined ratio. For a production analysis, consider keeping an explicit undefined/missing value or flag rather than treating the two cases as identical.

Key findings. The overall churn rate remains approximately 17.7%. Numeric activity variables show substantial spread and right-skew. Device-level churn proportions appear broadly similar. Driving-day frequency and recent activity are useful candidates for further investigation, but the visualizations alone cannot show that low activity causes churn.

Interpretation. EDA helps surface patterns and data issues, but results depend on preprocessing choices such as 95th-percentile capping. Any operational intervention should be evaluated with further analysis or an experiment.

5. Notebook 3 — The Power of Statistics

Purpose. Test whether mean drive counts differ between iPhone and Android users.

Method. The notebook maps device categories to numeric codes for group comparison, computes mean drives, and performs an independent two-sample Welch t-test (`equal_var=False`). The null hypothesis is that the population mean number of drives is equal across the two device groups; the alternative is that the means differ. The significance threshold is $\alpha = 0.05$.

Measure	Result
Mean drives — iPhone	67.86
Mean drives — Android	66.23
Welch t statistic	1.464
p-value	0.1434
Decision at $\alpha = 0.05$	Fail to reject H■; no statistically significant difference detected

Interpretation. Although the sample mean is slightly higher for iPhone users, p = 0.1434 exceeds 0.05. The analysis therefore does not provide sufficient evidence of a population mean difference at the selected significance level. This does not prove the means are identical, and it does not test whether device type causes churn.

6. Overall conclusions and recommended next steps

The three notebooks provide a coherent introductory workflow: inspect and understand the data, explore patterns visually, and use a formal test to assess one specific group difference.

Recommended next steps:

• Examine early engagement indicators such as activity days, driving days, and changes in recent sessions.
• Handle missing churn labels explicitly; do not automatically classify them as churned or retained.
• Revisit ratios with zero denominators and consider an explicit undefined-value flag.
• Document the 95th-percentile capping decision and compare results with uncapped data.
• Explore additional hypotheses with a clear business question, appropriate test assumptions, and consideration of multiple testing.
• If developing a churn model, use a defined target, appropriate train/test separation, baseline comparisons, and validation metrics.

7. Limitations and portfolio transparency

These are course-guided learning exercises, not a production churn model. The notebooks do not document a public data source or collection method; no causal claims are justified from these analyses. The 700 missing target labels, zero-denominator ratios, skewed distributions, and outlier capping all affect interpretation. The device hypothesis test concerns average drive counts, not churn directly.

The original notebooks contain Coursera lab instructions and course structure. Before making them public, review the applicable course-specific sharing and assessment rules. A portfolio can instead publish an independently written project overview, methodology, findings, and visualizations that you are permitted to share, with clear attribution to the learning context.

Prepared from the three supplied notebooks: Course 1 Waze Project Lab, Course 2 Waze project lab, and Course 3 Waze project lab. No outside data source has been inferred.